

Chap.1 Circuit Variables

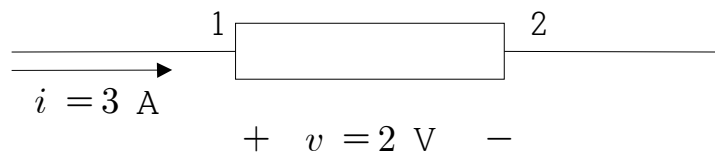
§1.4 Voltage and Current

- Voltage and Current defined in Physics

definition of voltage	definition of current
$v = \frac{dw}{dq}$	$i = \frac{dq}{dt}$
v = the voltage in volts w = the energy in joules q = the charge in coulombs	i = the current in amperes q = the charge in coulombs t = the time in seconds

§1.5 The Ideal Basic Circuit Element

(example) Interpretation of reference directions



2 volt voltage drop from terminal 1 to terminal 2
 or 2 volt voltage rise from terminal 2 to terminal 1

※ Power $p = vi = 2[\text{V}] \times 3[\text{A}] = 6[\text{W}]$ [W: Watt]

6 watts are consumed by the element. (heat)

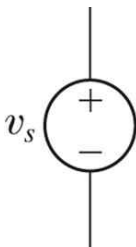
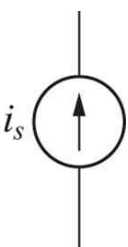
※ $p > 0$: power consumption by the circuit element
 (example: resistors)

$p < 0$: power generation by the circuit element
 (example: batteries)

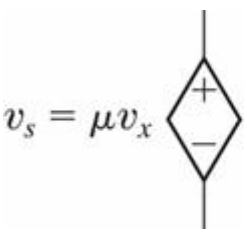
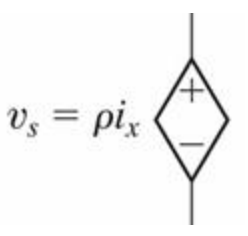
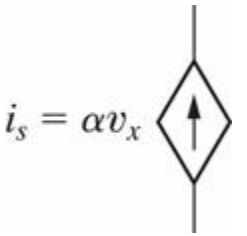
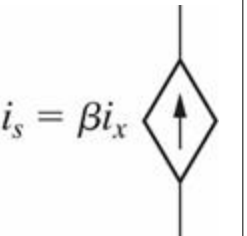
Chap.2 Circuit Elements

§2.1 Voltage Sources and Current Sources

- Ideal Independent Source

ideal independent voltage source	ideal independent current source
	

- Ideal Dependent Source

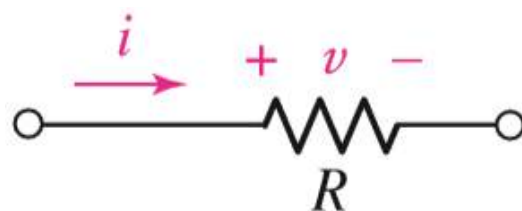
an ideal dependent voltage-controlled voltage source (VCVS)	an ideal dependent current-controlled voltage source (CCVS)	an ideal dependent voltage-controlled current source (VCCS)	an ideal dependent current-controlled current source (CCCS)
			

- active elements : can generate electric energy
- passive elements : cannot generate electric energy

§2.2 Electrical Resistance (Ohm's Law)

- Ohm's Law

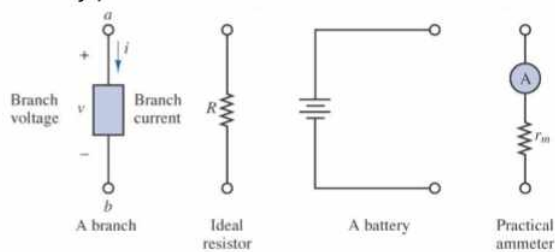
$$v = i R \quad [R = \text{the resistance in Ohms } (\Omega)]$$



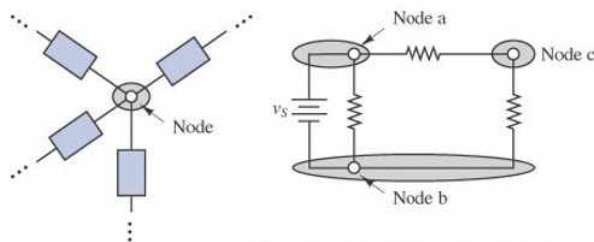
● Terminology

Branch & Node : Basic Units in the Circuit

- Branch : any portion of a circuit with two terminals

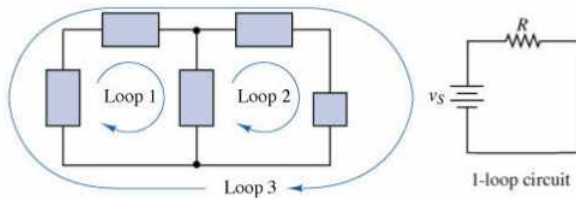


- Node : junction of two or more branches

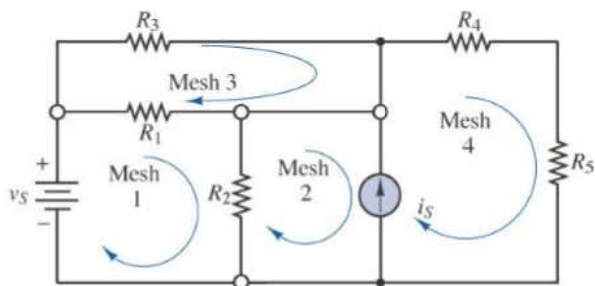


Loop & Mesh : Basic Units in the Circuit

- Loop : any closed connection of branches

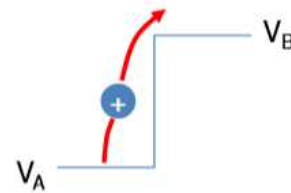


- Mesh : a loop that does not contain other loops



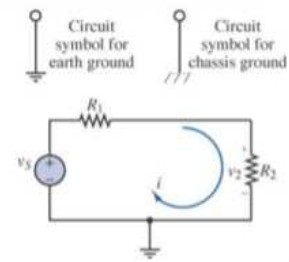
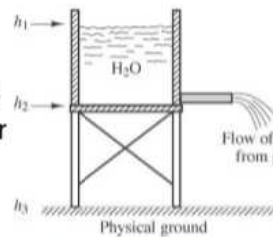
Voltage & Ground

- Voltage (Potential difference) :
 - The energy required to move charge from one point to the other.
 - Higher voltage means that more energy is required to move charge.
 - Unit : Volt (= Joule / Coulomb)



- Reference Voltage:
 - In a circuit, any one node may be chosen as the reference node, such that all node voltages may be referenced to this reference voltage.

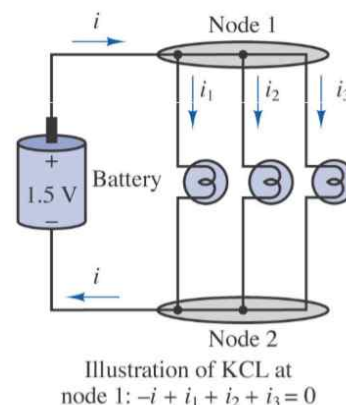
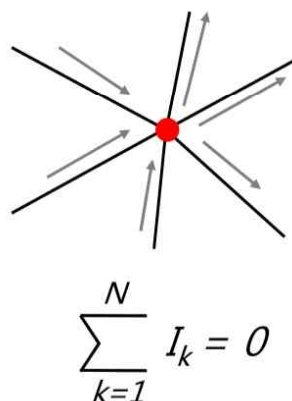
- Ground (voltage):
 - Reference point in an electrical circuit from which other voltages are measured, a common return path for electric current, or a direct physical connection to the earth.



§2.4 Kirchhoff's Laws

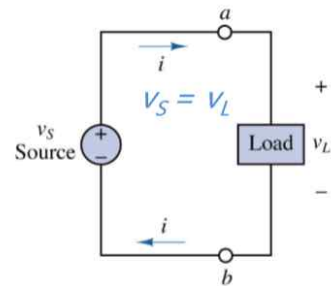
● Kirchhoff's Current Law (KCL)

- No charge can be created (except for current or voltage sources) in the electrical circuit;
- The sum of the currents at a node must equal zero.
- In other words, sum of "incoming" currents equals that of "outgoing" currents.

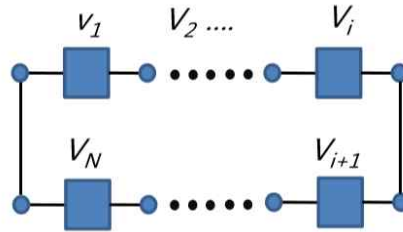


● Kirchhoff's Voltage Law (KVL)

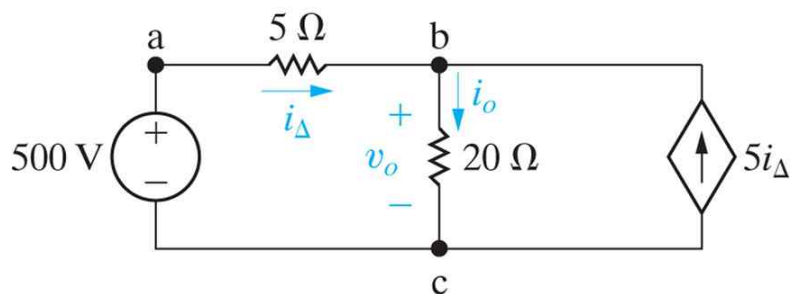
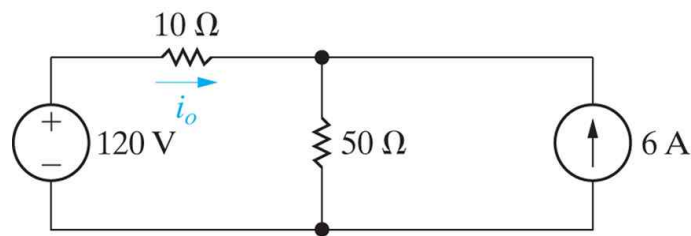
- Kirchhoff's Voltage Law (KVL) :
 - Net voltage around a close circuit is zero.
 - In other words, the sum of all voltages associated with sources equal that of load voltages.
 - In equation,



$$\sum_{k=1}^N v_k = 0$$



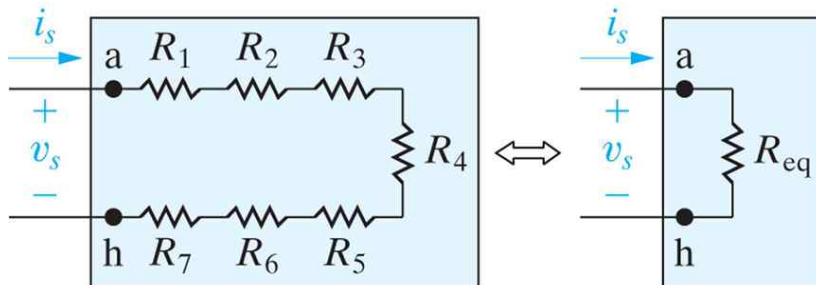
(Example) Applying Ohm's Law, KVL and KCL



Chap.3 Simple Resistive Circuits

§3.1 Resistors in Series

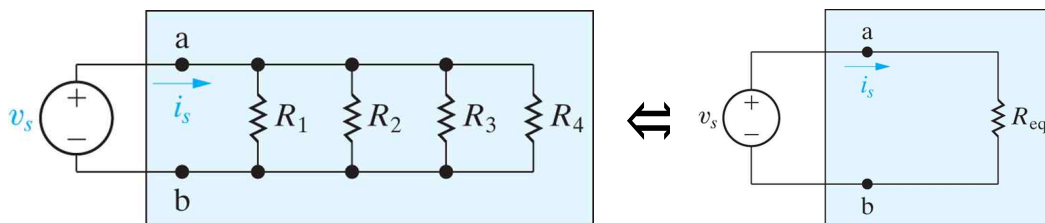
- Combining resistors in series



$$R_{eq} = \sum_{n=1}^N R_n = R_1 + R_2 + \dots + R_N$$

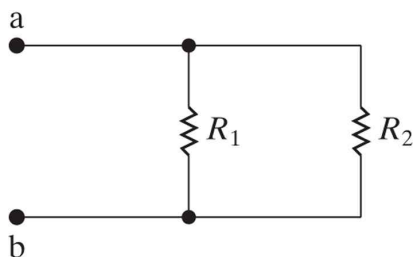
§3.2 Resistors in Parallel

- Combining resistors in parallel



$$\frac{1}{R_{eq}} = \sum_{n=1}^N \frac{1}{R_n} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}$$

- Two resistors in parallel (often used)



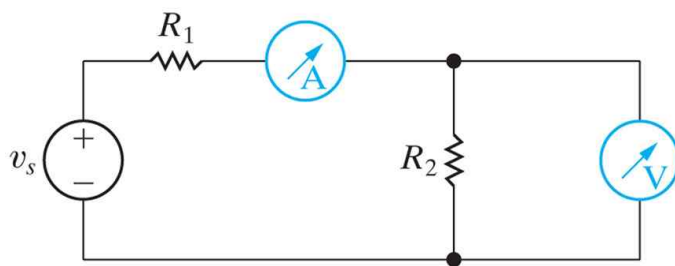
$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

§3.5 Measuring Voltage and Current

- measuring voltage [voltmeter] and current [ammeter]

an ammeter connected in series
to measure the current in R_1

a voltmeter connected in parallel
to measure the voltage across R_2

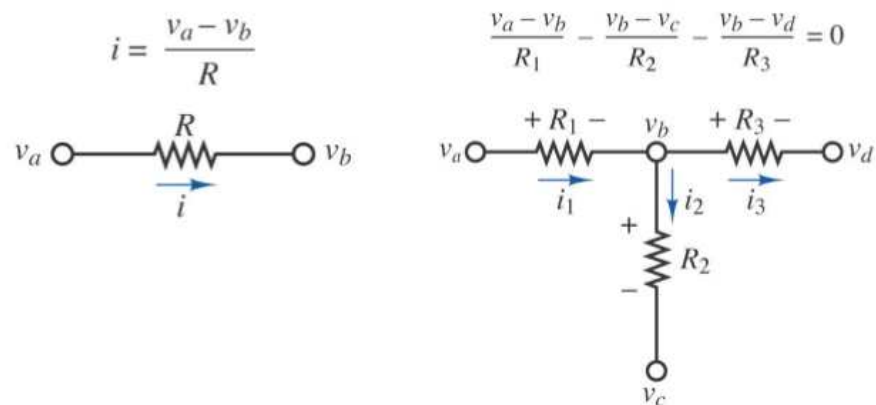


Chap.4 Techniques of Circuit Analysis

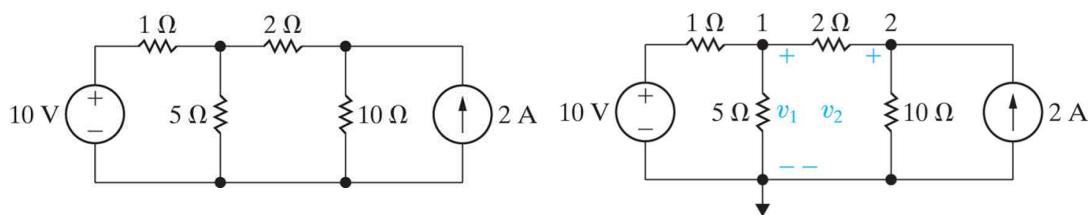
§4-2 Introduction to the Node-Voltage Method

Node Voltage Method

- Select one of the node as the **reference node** (usually ground).
- Define the voltage at each node as "independent variables".
- Use the Ohm's law to determine the current along the branch between nodes.
- Apply **Kirchoff's Current Law (KCL)** at each node.



(Example)

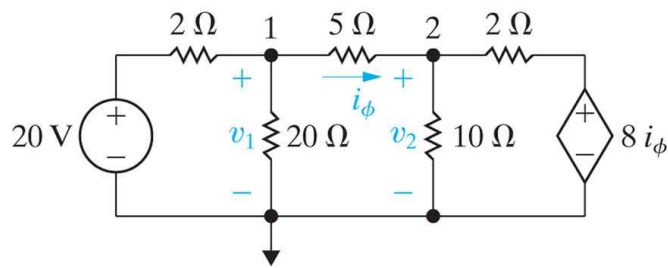


$$\text{node 1 : } \frac{v_1 - 10}{1} + \frac{v_1}{5} + \frac{v_1 - v_2}{2} = 0$$

$$\text{node 2 : } \frac{v_2 - v_1}{2} + \frac{v_2}{10} - 2 = 0$$

$$\therefore v_1 = \frac{100}{11} = 9.09 \text{ V}, \quad v_2 = \frac{120}{11} = 10.91 \text{ V}$$

(Example) The node-voltage method with dependent sources



$$\text{node 1 : } \frac{v_1 - 20}{2} + \frac{v_1}{20} + \frac{v_1 - v_2}{5} = 0$$

$$\text{node 2 : } \frac{v_2 - v_1}{5} + \frac{v_2}{10} + \frac{v_2 - 8i_\phi}{2} = 0$$

unknowns: 3. We need one more equation.

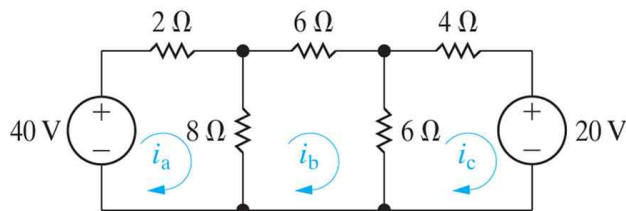
$$\text{3rd equation : } i_\phi = \frac{v_1 - v_2}{5}$$

$$\therefore v_1 = 16 \text{ V}, \quad v_2 = 10 \text{ V}, \quad i_\phi = 1.2 \text{ A}$$

§4.5 Introduction to the Mesh-Current Method

(※ It works only with PLANAR circuits.)

(Example) Using the mesh-current method



$$-40 + 2i_a + 8(i_a - i_b) = 0$$

$$8(i_b - i_a) + 6i_b + 6(i_b - i_c) = 0$$

$$6(i_c - i_b) + 4i_c + 20 = 0$$

Finally,

$$(2+8)i_a - 8i_b = 40$$

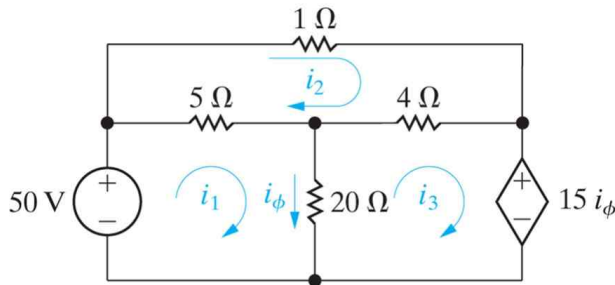
$$-8i_a + (8+6+6)i_b - 6i_c = 0$$

$$-6i_b + (6+4)i_c = -20$$

we can represent those equations using matrix

$$\begin{bmatrix} 10 & -8 & 0 \\ -8 & 20 & -6 \\ 0 & -6 & 10 \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = \begin{bmatrix} 40 \\ 0 \\ -20 \end{bmatrix}$$

(Example) the mesh-current method with dependent sources



$$50 = 5(i_1 - i_2) + 20(i_1 - i_3)$$

$$0 = 5(i_2 - i_1) + 1i_2 + 4(i_2 - i_3)$$

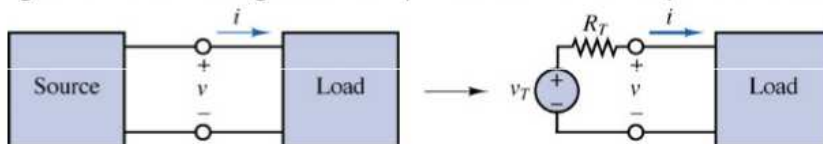
$$0 = 20(i_3 - i_1) + 4(i_3 - i_2) + 15i_\phi$$

$$i_\phi = i_1 - i_3$$

§4.10 Thevenin and Norton Equivalent Circuits

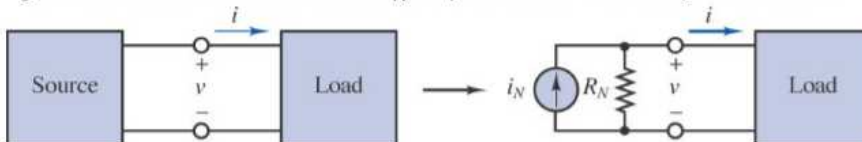
- Thevenin Theorem

When viewed from the load, any network composed of ideal voltage and current sources, and of linear resistors, may be represented by an equivalent circuit consisting of an ideal voltage source v_T in series with an equivalent resistance R_T .



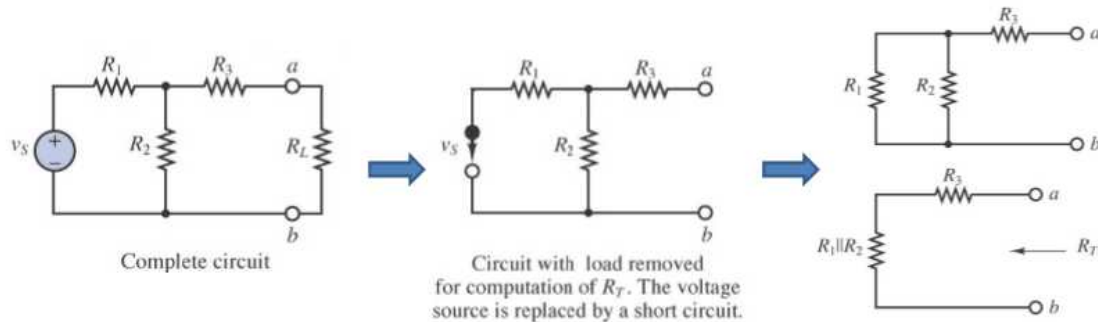
- Norton Theorem

When viewed from the load, any network composed of ideal voltage and current sources, and of linear resistors, may be represented by an equivalent circuit consisting of an ideal current source i_N in parallel with an equivalent resistance R_N .



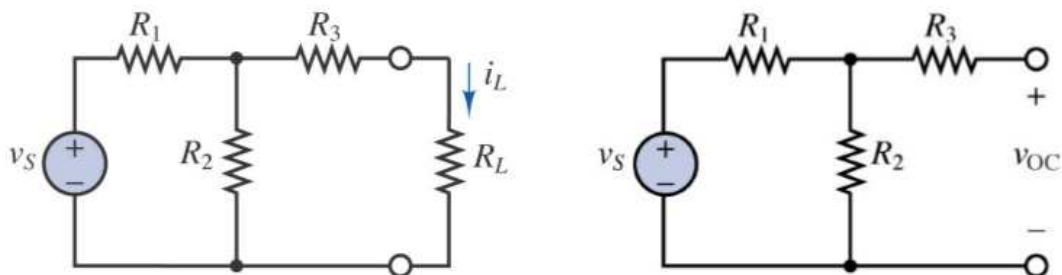
How to Determine Norton or Thevenin Equivalent Resistance?

- The Thevenin and Norton equivalent resistances are one and the same quantity.
- We obtain it by the following steps:
 1. Remove the load
 2. Zero all independent voltage and current sources.
 3. Compute total resistance between load terminals with the load removed.



How to Compute the Thevenin Voltage?

- The equivalent (Thevenin) source voltage is equal to the "**open-circuit voltage**" present at the load terminals with the load removed.
- Procedure:
 1. Remove the load (load terminal: open circuited)
 2. Apply either node voltage method or mesh current method to solve the open circuit voltage (v_{oc}).
 3. The Thevenin voltage is $v_T = v_{oc}$.



How to Compute the Norton Current?

- The equivalent (Norton) source current is equal to the "short-circuit current" that would flow if the load were replaced by a short circuit.

- Procedure:

1. Replace the load with a short circuit.
2. Apply either node voltage method or mesh current method to solve the short circuit current (i_{sc}).
3. The Norton current is $i_N = i_{sc}$.

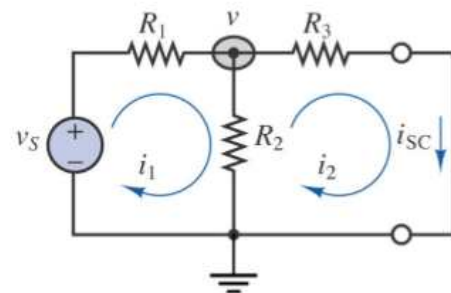
Mesh 1: $v_s - R_1 i_1 - R_2 (i_1 - i_2) = 0$

Mesh 2: $-R_2 (i_2 - i_1) - R_3 i_2 = 0$



Compute the current i_2 .
Then,

$$i_N = i_{sc} = i_2$$



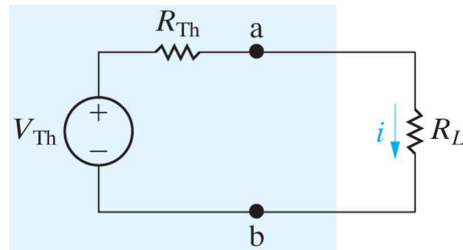
● Summary: How to get Thevenin and Norton equivalent circuits

case	Network A	Thevenin & Norton
I	independent sources and resistors	Find two of v_{oc} , i_{sc} , and R_{TH} . Calculate one by $v_{oc} = R_{TH} i_{sc}$.
II	independent sources, dependent sources and resistors	Find v_{oc} and i_{sc} . Calculate R_{TH} by $R_{TH} = v_{oc} / i_{sc}$.

(※ $v_{oc} = v_{TH}$ in the Thevenin equivalent circuits)

(※ $i_{sc} = i_N$ in the Norton equivalent circuits)

§4.12 Maximum Power Transfer



$$p_L = i^2 R_L = \left(\frac{V_{TH}}{R_{TH} + R_L} \right)^2 R_L$$

Find the maximum value of p_L

$$\frac{dp_L}{dR_L} = V_{TH}^2 \left[\frac{(R_{TH} + R_L)^2 - R_L \cdot 2(R_{TH} + R_L)}{(R_{TH} + R_L)^4} \right] = 0$$

$$\therefore (R_{TH} + R_L)^2 - R_L \cdot 2(R_{TH} + R_L) = 0$$

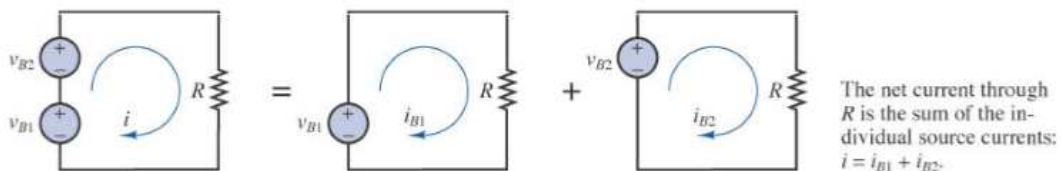
$$\therefore R_L = R_{TH}$$

(※ When $R_L = R_{TH}$, the power consumed at R_L is maximum.

$$p_{L \max} = \frac{V_{TH}^2 R_L}{(2R_L)^2} = \frac{V_{TH}^2}{4R_L}$$

§4.13 The Principle of Superposition

- In **LINEAR*** circuit that contains N sources, each branch voltage and current is **the sum of N voltages and currents** (i.e. sum of the results from N independent sources).
- Each set of these voltages and currents can be computed by **setting only one (voltage or current) source ON and all others ZERO** (i.e. OFF) and solving the circuit containing that single source.

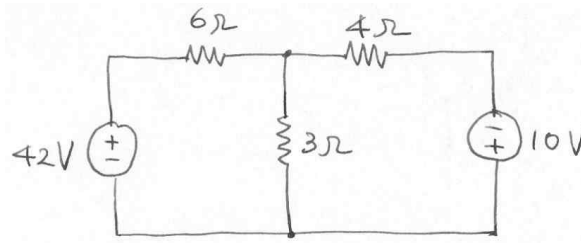


* Linearity of system

- Additivity : $f(x+y) = f(x) + f(y)$
- Homogeneity : $f(\alpha x) = \alpha f(x)$

※ Superposition is a very important concept for a linear system, but superposition principle is not widely used to analyze a multi-source circuit, since we need to analyze the circuit several times.

(Example) Find the voltage across the 3Ω resistor. [answer: 6V]

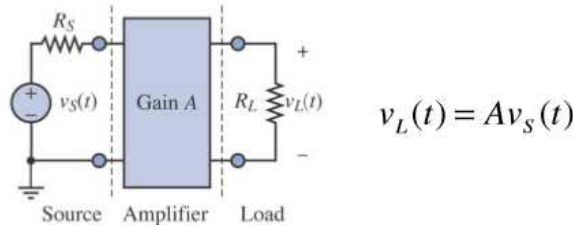


- (1) Ohm's Law, KVL, KCL
- (2) Node-voltage method
- (3) mesh-current method
- (4) superposition method
- (5) source transformation
- (6) Thevenin (or Norton) equivalent circuit

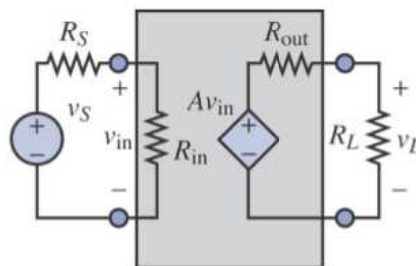
Chap.5 The Operational Amplifier

§5.0 Ideal Amplifiers

- Ideal Amplifiers: amplification of input signal to output signal with a certain amplification ratio.



- The internal structure of the amplifier can also be represented with Thevenin's equivalent circuit.

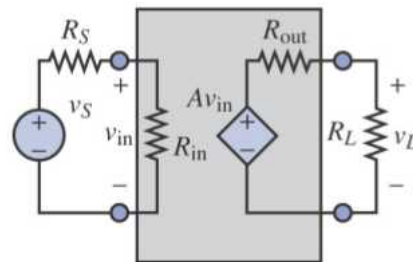


Ideal Amplifiers

$$v_{in} = \frac{R_{in}}{R_S + R_{in}} v_S$$

$$v_L = A v_{in} \frac{R_L}{R_L + R_{out}}$$

$$= \left(A \frac{R_L}{R_L + R_{out}} \frac{R_{in}}{R_S + R_{in}} \right) v_S$$

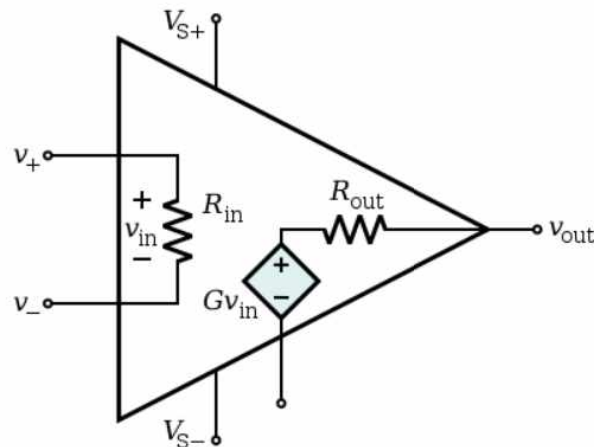


- ➡ • The amplification factor is dependent on both the source and load impedance
- If the input resistance R_{in} is very large & output resistance R_{out} is very small,

$$v_L \approx \left(A \frac{R_L}{R_L} \cdot \frac{R_{in}}{R_{in}} \right) v_S \approx A v_S$$

- ➡ Amplification factor is independent of source and load impedances.

- Ideal operational amplifier



1. Ideal op amp behaves as an ideal difference amplifier (amplifies the difference between two input voltages).

$$v_0 = A(v_p - v_n)$$

2. near-infinite input resistance

$$R_{in} = \infty$$

3. near-zero output resistance

$$R_{out} = 0$$

4. Due to near-infinite input resistance, it is usually assumed that the input current is zero.

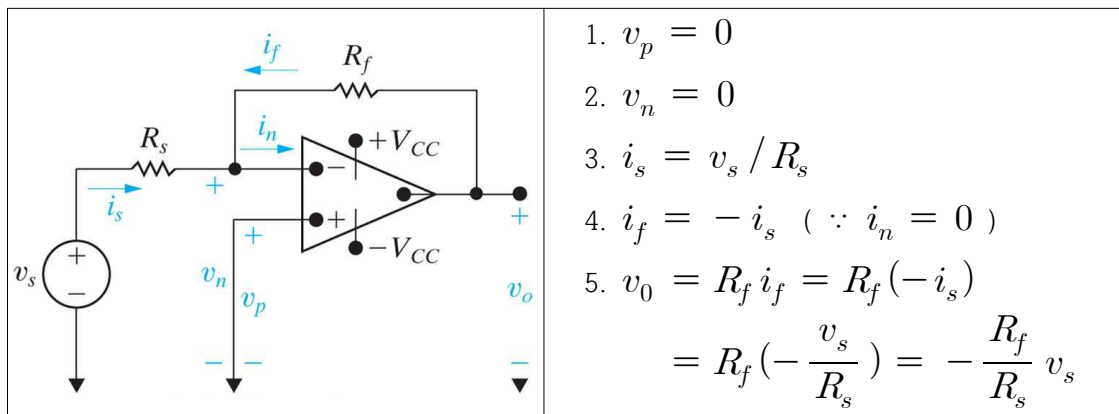
$$i_p = i_n = 0$$

5. Due to near-infinite open-loop gain ($A = \infty$)

$$v_p = v_n$$

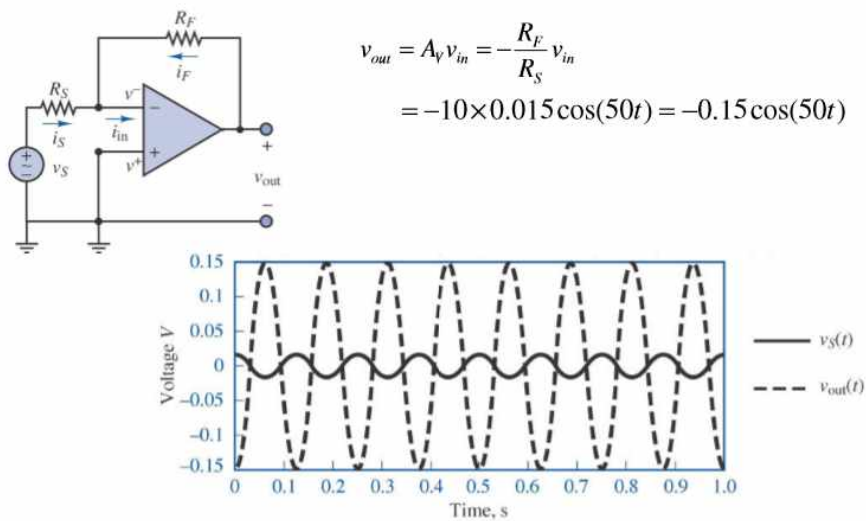
(※ actual op amp : $A = 10^5 \sim 10^7$)

§5.3 The Inverting-Amplifier Circuit



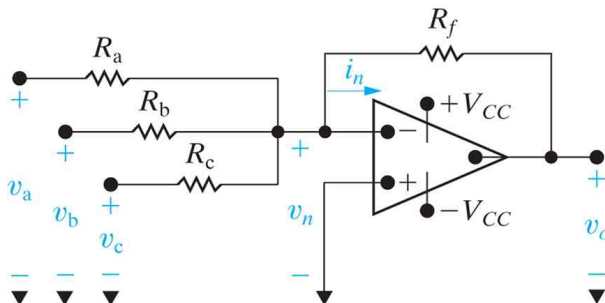
(Example) Inverting Amplifier

• If $R_s = 1k\Omega$, $R_f = 10k\Omega$, and $v_s = 0.015\cos(50t)$



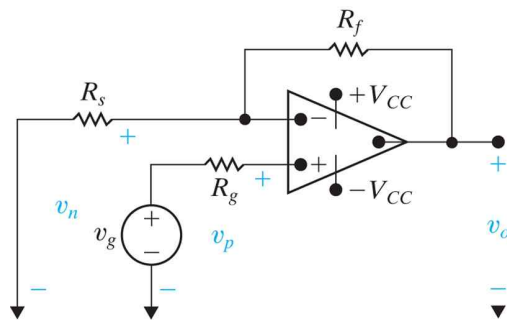
§5.4 The Summing-Amplifier Circuit

[※ Adder in analog systems]



$$v_o = - \left(\frac{R_f}{R_a} v_a + \frac{R_f}{R_b} v_b + \frac{R_f}{R_c} v_c \right)$$

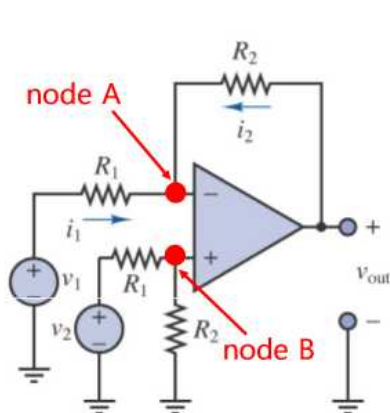
§5.5 The Noninverting-Amplifier Circuit



$$v_o = \frac{R_s + R_f}{R_s} v_s = \left(1 + \frac{R_f}{R_s}\right) v_s$$

(* Not widely used.)

§5.6 The Difference Amplifier Circuit



- KCL at node A,

$$i_1 + i_2 = \frac{v_1 - v^-}{R_1} + \frac{v_{out} - v^-}{R_2} = \frac{v_1 - v^+}{R_1} + \frac{v_{out} - v^+}{R_2} = 0$$

- At node B,

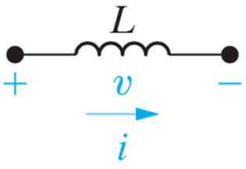
$$v^+ = \frac{R_2}{R_1 + R_2} v_2$$

$$\begin{aligned} \Rightarrow \frac{v_{out}}{R_2} &= \left(\frac{1}{R_1} + \frac{1}{R_2}\right) v^+ - \frac{1}{R_1} v_1 \\ &= \left(\frac{1}{R_1} + \frac{1}{R_2}\right) \frac{R_2}{R_1 + R_2} v_2 - \frac{1}{R_1} v_1 = \frac{1}{R_1} (v_2 - v_1) \end{aligned}$$

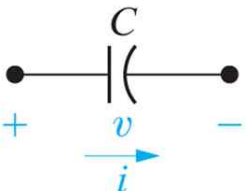
$$\Rightarrow \boxed{v_{out} = \frac{R_2}{R_1} (v_2 - v_1)} \quad \text{Differential amplifier closed-loop gain}$$

Chap.6 Inductance and Capacitance

§6.1 The Inductor

 L : Henry [H]	$v(t) = L \frac{di(t)}{dt}$ $i(t) = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$ Power : $p(t) = v(t) i(t) = L i(t) \frac{di(t)}{dt}$ Energy : $w(t) = \frac{1}{2} L i^2(t)$
---	--

§6.2 The Capacitor

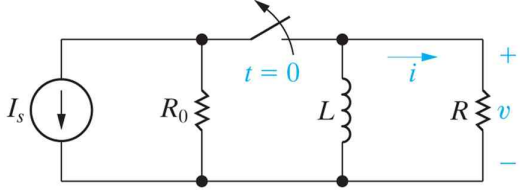
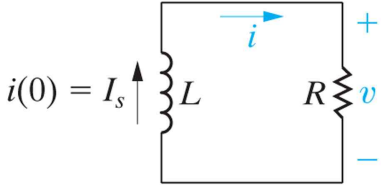
 C : Farad [F]	$i(t) = C \frac{dv(t)}{dt}$ $v(t) = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$ Power : $p(t) = v(t) i(t) = C v(t) \frac{dv(t)}{dt}$ Energy : $w(t) = \frac{1}{2} C v^2(t)$
---	--

● Continuity

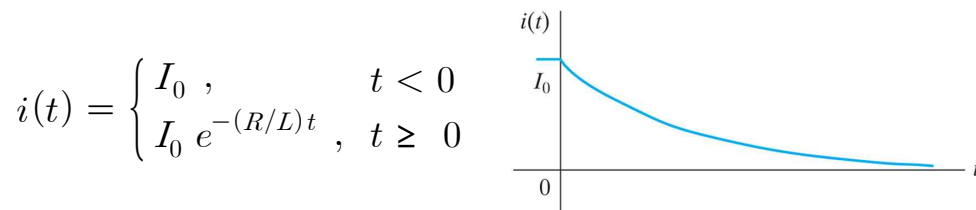
Inductor	Capacitor
$v_L(t) = L \frac{di_L(t)}{dt}$	$i_C(t) = C \frac{dv_C(t)}{dt}$
$i_L(0^-) = i_L(0) = i_L(0^+)$	$v_C(0^-) = v_C(0) = v_C(0^+)$
$i_L(t)$: always continuous $v_L(t)$: may or may not	$v_C(t)$: always continuous $i_C(t)$: may or may not

Chap.7 Response of First-Order RL and RC Circuits

§7.1 The natural Response of an RL Circuit

an RL circuit with a switch	the circuit for $t \geq 0$
	
KVL (differential equation) [* homogeneous differential eq.] $L \frac{di(t)}{dt} + Ri(t) = 0 \quad (\text{initial condition: } i(0) = I_s \equiv I_0)$ solving the differential equation $i(t) = i(0) e^{-(R/L)t}, \quad t \geq 0$ since the current in an inductor must be continuous $i(0^-) = i(0) = i(0^+) = I_0$ $\therefore$ natural response : $i(t) = I_0 e^{-(R/L)t}, \quad t \geq 0$	

- the natural response of an RL circuit



- the significance of the time constant

When $e^{-(R/L)t} = e^{-1} = 0.36788 \approx 0.37$,
the time t is the TIME CONSTANT, τ .

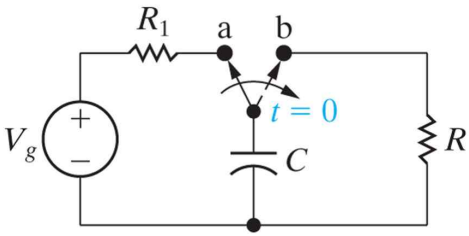
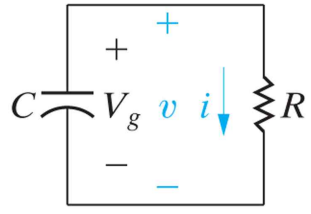
$$\therefore \tau = \frac{L}{R} \quad [\text{sec}]$$

t	$e^{-t/\tau}$
τ	0.3678794
2τ	0.1353353
3τ	0.0497871
4τ	0.0183156
5τ	0.0067379

※ transient response

※ steady-state response

§7.2 The natural Response of an RC Circuit

an RC circuit with a switch	the circuit for $t \geq 0$
	
KCL [* homogeneous differential eq.] $C \frac{dv(t)}{dt} + \frac{v(t)}{R} = 0 \quad (\text{initial condition: } v(0) = V_g \equiv V_0)$ Solving this 1st order differential equation $v(t) = v(0) e^{-t/RC}, \quad t \geq 0$ Since the voltage across a capacitor is continuous $v(0^-) = v(0) = v(0^+) = V_0 = V_g$ $\therefore$ natural response : $v(t) = V_0 e^{-t/RC}, \quad t \geq 0$	

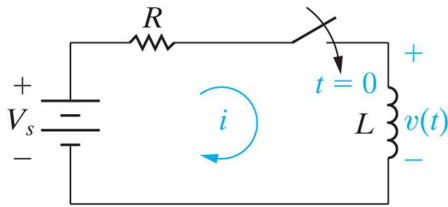
- the time constant τ for an RC circuit

Since $v(t) = V_0 e^{-t/RC}, \quad t \geq 0,$

the time constant τ of RC circuit is $\tau = RC$ [sec]. (*important)

§7.3 The Step Response of RL and RC Circuits

● the step response of an RL circuit



$$\text{KVL: } L \frac{di(t)}{dt} + Ri(t) = V_s$$

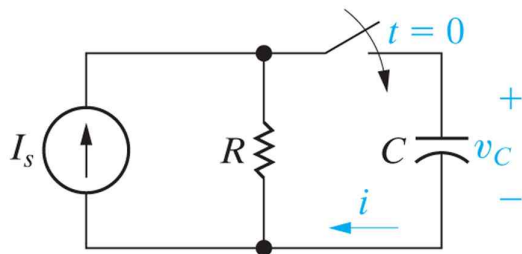
[* nonhomogeneous differential equation]

Solving the differential equation, the solution is;

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R} \right) e^{-(R/L)t}, \quad t \geq 0$$

$$= i(\infty) + (i(0) - i(\infty)) e^{-(R/L)t}$$

● the step response of an RC circuit



$$\text{KCL: } C \frac{dv_C(t)}{dt} + \frac{v_C(t)}{R} = I_s$$

[* nonhomogeneous differential equation]

Solving the differential equation, the solution is

$$v_C(t) = I_s R + \left(V_0 - I_s R \right) e^{-t/RC}, \quad t \geq 0$$

$$= v_C(\infty) + (v_C(0) - v_C(\infty)) e^{-t/RC}$$

§7.4 A General Solution for Step and Natural Responses

- general solution for natural and step responses of RL/RC circuits

$$x(t) = x_f + (x_0 - x_f) e^{-t/\tau}$$

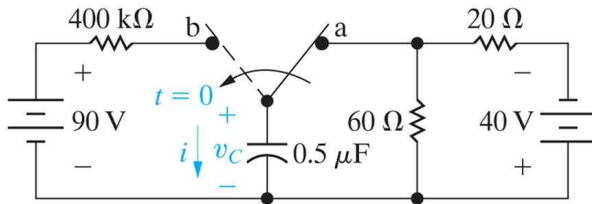
$x(t)$: the unknown variable as a function of time

x_f : the final value of the variable

x_0 : the initial value of the variable

τ : the time constant

(Example)



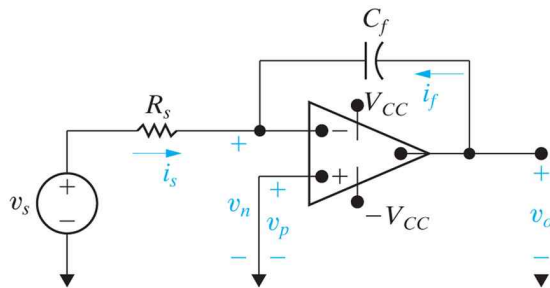
(i) the initial value of $v_C(t)$: $v_C(0^-) = v_C(0^+) = -30 \text{ V}$

(ii) the final value of $v_C(t)$: $v(\infty) = +90 \text{ V}$ (i.e., v_f)

(iii) the time constant τ : $\tau = RC = (400\text{k}) \times (0.5\mu\text{F}) = 0.2 \text{ s}$

(iv) $v_C(t) = 90 + (-30 - 90) e^{-5t} = 90 - 120 e^{-5t} \text{ V}, \quad t \geq 0$

§7.7 The Integrating Amplifier



$$v_o(t) = -\frac{1}{R_s C_f} \int_{t_0}^t v_s(\lambda) d\lambda + v_o(t_0)$$

※ If we use this circuit, we can solve a first-order differential equation. That is, we can make an analog computer.